



Society of Petroleum Engineers

SPE-229278-MS

Digital Replica of HPHT Inter-Field Flowlines

Roberto Bruschi and Giovanni Buizza Avanzini, DNV, Milan, Italy; Danilo Machado L da Silva, DNV, Lisbon, Portugal

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229278-MS](https://doi.org/10.2118/229278-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

A subsea pipeline network is part of the complex field development infrastructure, which includes different sub-systems, from surface to seabed. The subsea pipeline network is a pivotal component, from well-head to manifold, then to export line or riser base. The satisfactory performance in operation, in both shallow and deep waters, depends on flow assurance and pipeline integrity. High pressure high temperature (HPHT) aggressive fluids from wells, often multi-phase, are challenging. At design stage, this is commonly tackled by using engineering models to predict performance in operation. These are inter-linked, increasingly high-fidelity, possibly allowing for multi-physics simulation. Nowadays, field development is based on a combination of engineering models and data from experience through installation to field monitoring, with the aim of building a digital replica or twin of the system on a software platform, which use is assisted by data analytics.

The scope of the paper is to show how a digital replica (digital twin, DT) of the inter-field pipeline network, likely part of the digital replica of the full system, can positively impact both design choices and operational strategy. DT are guided by safety targets and operational performance to be met, the latter often input/output to/from other sub-systems. The paper deals with a new approach to subsea pipeline networks, belonging to complex field development infrastructures, which is based on extensive condition monitoring concept. There is a digital replica of the infrastructure, both functional and structural, conceived at the early design, improved during installation and pre-commissioning, continuously updated in operation. DT is unique to deal with complexity, allowing for testing flexibility and reliability of adopted solutions at design stage, with the aim to timely and rationally tackle those variations that might deeply affect life-cycle performance.

Introduction

When the infrastructure of offshore Oil & Gas (OG) production is complex, whether in shallow or deep waters, condition monitoring in operation becomes especially relevant. Overall, the operator's and owner's targets are to improve safety and be (cost) effective. Attaining the desirable performance depends, largely, on design choices undertaken to tackle the uncertainties that are always present at early design stage. And while shallow and deep waters applications are different, they share similar criticalities, namely flow assurance

and integrity of the pipelines. The intra-field pipeline network shall be designed accounting for chemistry (e.g. percentage of carbon dioxide or hydrogen sulfide causing sweet or sour corrosion of line pipe steel) and physics (e.g. high-pressure and high-temperature – HPHT – flow causing thermal expansion of the steel pipeline on the seabed) of fluids from well-head. Aggressive HPHT flows might threaten the structural integrity of the pipelines in the long run. Design adopts a safe life approach; however, it might not always meet the challenge.

Digital twin for HPHT flowlines: why

At design stage, material performance is tested, flow regimes are defined, and pipeline response to load effects is calculated. However, uncertainties of parameters at design stage (chemistry and physics of flows), as well as changing operational needs during the life cycle (flow fluctuations and purpose change of their use), pose questions about the field performance of subsea pipeline network over the years. Safe-life design cannot always provide a satisfying answer, even when conservative and costly choices are made at design stage. The target is safe performance over the full life cycle. New technological advancements are applied to new developments of complex offshore OG fields (Zhang et al., 2019). A digital twin of the full infrastructure is here proposed, from early design stage, to optimize on paper the choices of components and their integration, then to follow operation through condition monitoring driven by data from sensors and surveys of physical model and simulation of the mirroring digital model. The aim is to properly tackle parameter uncertainty as well as variations from early design to end of life. Realistic virtual models of the system can bridge design and construction, by mirroring the virtual model and real world (Van et al., 2023). HPHT subsea infrastructure, often including different flowline materials (carbon steel, corrosion resistant alloys) and concepts (pipe cladding, burial and/or cover, wet insulation, pipe-in-pipe), is a typical system whose performance can take advantage of digital twin technology (Yang et al., 2023).

Historical perspective

The digital twin or replica of a pipeline is not a new concept per se: pipeline integrity management systems (PIMS) are its precursor (Amaechi et al. 2023). Already in the early days of strategic transcontinental gas pipelines, in the ‘80s, it was common to monitor real time the gas flow along the pipeline, to control and activate measures in case of gas leak. This has been further developed in the North Sea, to control in real time the blending of natural gas transported to central Europe from different production fields (Helgaker and Ytrehus, 2012). Data from regular surveys were gathered and processed, without real time monitoring. This was mainly related to the peculiarity of natural gas transportation: limited corrosion, plastic strength resources, damage tolerance of line pipe steel lead to consider real time monitoring as superfluous. Things are changing nowadays. Turbulent geopolitical scenarios and technological advancements allow for sabotage, as shown by recent accidental events (NATO, 2023). Real time intrusion monitoring is becoming a new area of potential development for PIMS. Further, the move into deeper water in the late ‘90s led to demanding HPHT productions, aggressive for the traditional C-Mn steel of pipelines and risers (Shadravan and Amani, 2012). R&D provided design methods and technological solutions, nonetheless the issue is still, in mid-2025, on the desk of Operators, OG Companies, Investor, Insurance Groups, and National Authorities as well. The digital transformation is fostering new development programs, such as digital twins.

Digital transformation

During the last decade, after a deep crisis of offshore OG industry initiatives, a new season has started. After the introduction of the brand-new concept of the subsea factory (Rognø, 2014), automation has become a staple, based on experience over years of operation and enabled by digital transformation advancement. Using data, often already available but unleveraged, and now increasingly abundant and reliable, is the new deal. Big data, communication, data analytics coupled with multi-scale high-fidelity modelling are the pillars, grounded on the new computer science. Artificial neural networks use data in

combination with traditional modelling to improve engineering prediction. Machine learning uses historical and increasingly available data by new and cheap instrumentation to monitor the performance in the field, and to improve asset management aiming at consistent condition predictions. Artificial intelligence assists operations, providing warning and contributing to decisions. This is leading to what can be defined as "data-assisted" engineering, or even "data-driven" (Augustyn et al., 2019). Offshore OG industry, ever receptive and inclined to innovation, is dedicating massive investments to this new promising data-driven sector, abandoning the temporary pause in the development of advanced subsea solutions that characterized the last decade. Time, however, was not lost in this period: new attention was dedicated to old infrastructures, in shallow waters, either for upgrading them with the aim of extending life, or for decommissioning. For the former, software platforms that can refer to project data and survey outcomes, when rational decisions are needed to safely continue operations, have been relevant. The more complex the infrastructure, the more demanding its asset integrity management system (PIMS for the pipeline networks). Fig. 1 shows the integration between subsea field, software platform storing project databases and survey results, monitoring system transferring data to the production team, expert network providing advice any time anomalies are detected (Bruschi, 2017).

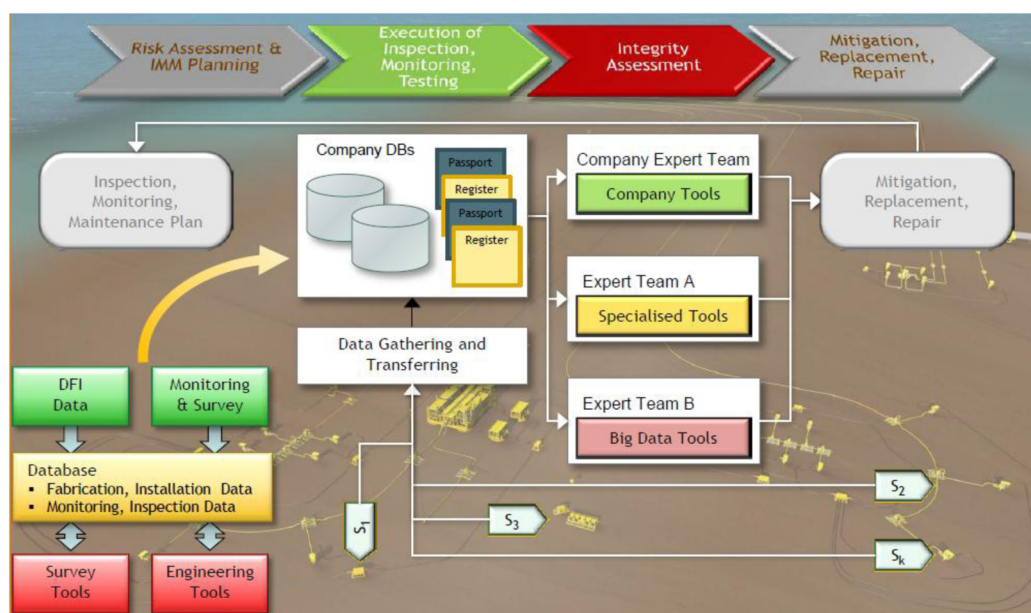


Figure 1—The desirable world of subsea flowline network management that is increasingly actual worldwide, in recent and near-to-come field develop projects, across both traditional offshore districts and new ultra deep-water ventures. Data communication is a factor.

From a safe life to fail-safe approach

Threats regarding subsea pipeline infrastructure are commonly related to corrosion, often unpredictable due to the variability of flow conditions (in contrast e.g. to natural gas transportation pipelines). In certain shallow water offshore districts, and typically in deep waters, the combination of corrosion and HPHT fluid are concerns for the structural integrity of the pipelines. HPHT flows can in fact modify the initial alignment of flowlines, possibly provoking an undesirable localization of strains (in service buckling). It is therefore common to perform post-design assessments during early service, to verify the configuration after start-up and/or activate remedial measures, if needed. The common practice is safe-life design, i.e. design choices that would allow for safe operation over lifespan, without any specific intervention. These commonly imply high investment costs, but not necessarily safe solutions. From this consideration, it becomes clear that material strength degradation cannot be fully tackled at design stage, only. Besides, costly pipeline concepts, or new material choices, taken at design stage to mitigate risk, might still not be enough to guarantee a

safe life. It is then necessary to rationally monitor the fluctuations of HPHT flows of aggressive fluids during the asset lifespan. An approach that accounts for the management of uncertainties through condition monitoring and rationally based interventions during service can lead to a new concept: fail-safe design. It implies inspection and monitoring as an effective tool to control operations and intervene to solve any undesirable degradation. Fail-safe design was first introduced in the aircraft industry in the '70s. Reliability-based inspection can be a design parameter that guarantees the control of degradation mechanisms having slow propagation to failure. This approach, applied to pipelines, requires a deep knowledge of limit states that are relevant and specific for different applications. Over-sizing the wall thickness, injecting inhibitors for tackling the corrosion risk, selecting expensive corrosion resistant alloys, all can be reconsidered. This is condition monitoring, an effective alternative to overdesign.

Full speed ahead, but carefully

Experience shows that it is unusual that (transportation) pipelines need for real time response to manage ordinary service, except in case of destructive accidental interference. Material strength degradation takes time, even if it is difficult to predict how much and where it is critical. However, for aggressive HPHT flows, the development of a digital twin can be justified to rationally solve the uncertainties that affect design and operation. And, in case of complex subsea field for offshore production, digital twin is a need. As for any change, there is a consensus between parties to be reached, engineering guidelines to be developed, technological advancements in data gathering, communication and analytics to be improved. The potential to improve safety and reduce both investment and operating costs is significant.

Complexity, flexibility, safety and a digital replica from early design stage

In the new millennium, economics has led Offshore OG to target a better exploitation of submarine reservoirs. The goal, namely to almost double exploitation (from 30%-35% to 60%-70%), is very ambitious. Technological advances are pivotal to reaching this goal: big data, IoT, data analytics and new digital tools (artificial neural networks, machine learning, artificial intelligence). In new prospects, seismic data is regularly gathered from a network of cabled sensors that cover the footprint of the reservoir. Data feeds 4D software models of reservoirs that need super-computers to run. The aim of the 4D model for a specific site is to improve the control of exploitation. This is also achieved by regularly updating the reservoir model using data from production flows (extracted and injected), performing simulations to compare predicted and actual production, and providing indications to guide field operations through remotely controlled well-head and manifold valves. The picture is completed by resident autonomous vehicles that perform inspection and operations using vision technology and advanced manipulators, giving rise to a complex system, including functionally interacting and physically interfering sub-systems (Fig. 2).

Complexity, flexibility, safety and efficiency are attributes of field development, that are maintained from the early stage to the end of the life cycle. The very ambitious exploitation targets, in combination with massive investments and the implicit risks due to limited field experience of new technologies, lead to requesting capacity to simulate in the digital world the different angles and steps of field development, with confidence, from the early stage of design to the end of life of the infrastructure. This is the role that is presently undertaken and increasingly covered by digital replica of the system. Up to today, this was done via software simulation, both functional and structural, carried out at the design stage based on parameters from project design basis. Nowadays, simulation tools are evolving into digital twins of the system, deployed through a software platforms, covering functional (the flow) and structural performance in service (asset integrity). DTs are continuously fed by real time data from instrumentation, purpose designed and proven. Data are assured, analyzed and integrated, thus allowing the DT to provide a realistic status of the system, very close to the actual condition but without imposing frequent field inspection, measurements campaign and system tests.

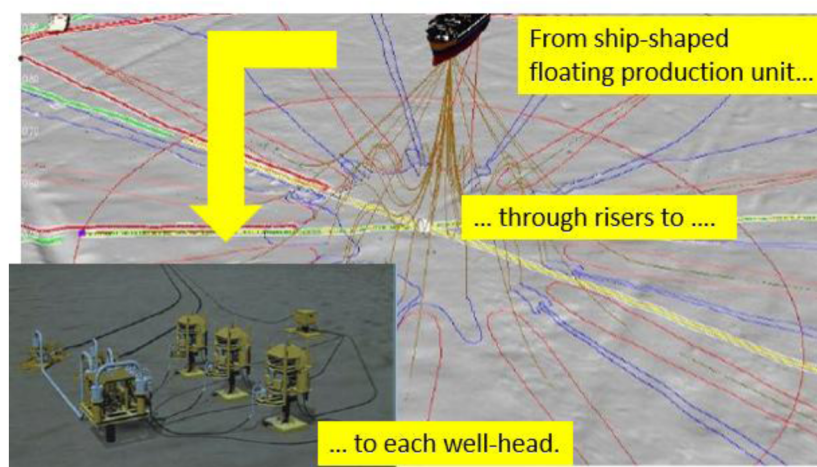


Figure 2—The complexity of ultra-deep water field development and the crowded subsea over wide footprint: from the ship-shaped floating production unit (FPU), moored to seabed at anchors, through production and export risers, dynamic umbilical's as well, to the subsea flowline and static umbilical network, that connect the well-heads through jumpers to manifolds then to the riser bases.

Creating a digital replica of this infrastructure can significantly enhance both the design and operational phases. During design, the asset model can be used to conduct a wide range of analyses that encompass the full uncertainty envelopes of design parameters (flow composition and regimes, soil characteristics, and material properties), enabling the selection of optimal configurations. Techniques like Monte Carlo simulations can support informed, risk-based decision-making. While such analyses are already performed to some extent, we propose expanding the scope of simulated conditions from the conceptual design phase onward. Models developed during the design phase are often neglected once operations begin. Our approach advocates for leveraging these models as the foundation for the asset's operational DT. These models must be continuously updated throughout the asset's lifecycle to reflect real-world conditions and modifications not captured by sensor systems. During operations, the DT is fed with live data to monitor degradation mechanisms, simulate scenarios, and predict performance, supporting full reservoir exploitation.

According to DNV-RP-A204, "a digital twin is a virtual representation of a system or asset that calculates system states and makes system information available, through integrated models and data, with the purpose of providing decision support over its life cycle." Fig. 3, taken from the recommended practice, shows the need for integration of project data, field outcomes and monitoring to get a tool that improves safety and performance. It emphasizes the importance of utilizing the DT throughout the asset's lifecycle to support a fail-safe engineering approach. Although the concept of using DT from the design phase is not new, its full implementation across the oil and gas industry remains limited.

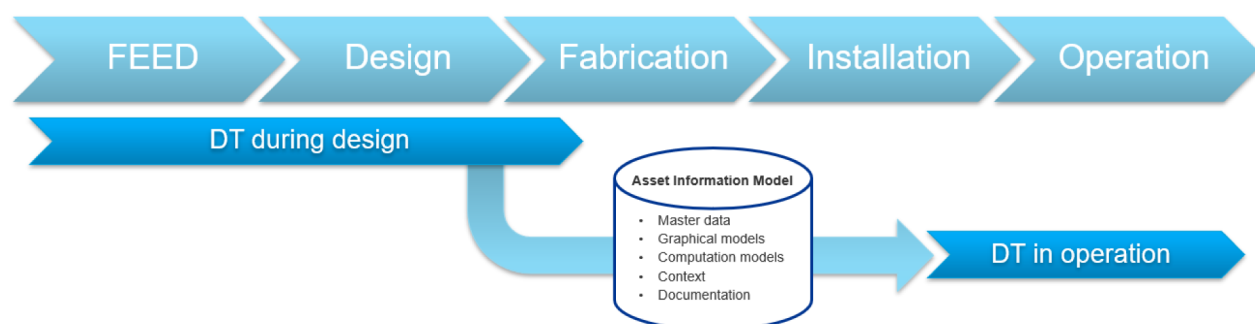


Figure 3—Connecting design to operations from digital twins, as taken from DNV-RP-A204, in a software platform where the physical system is mirrored by the digital model i.e. the digital twin.

Developing and maintaining a DT is a complex undertaking. Its justification depends on whether the system's complexity warrants a condition monitoring infrastructure over traditional approaches. The return on investment (ROI) must be evaluated in terms of potential savings in design, maintenance, and life extension assessments. This is more likely to be favorable for complex systems with multiple uncertainties than for simpler, tightly controlled infrastructures. Adoption also hinges on industry acceptance of fail-safe methodologies, regulatory frameworks, and the integration of digital twin workflows into organizational processes. As technology advances—making sensors more affordable, analytics more accessible, and data-driven practices more prevalent—the case for digital twins becomes increasingly compelling.

Subsea field development infrastructure is inherently complex, comprising multiple subsystems such as subsea trees, manifolds, jumpers, flowlines. The new concepts of field development in remote regions and ultra-deep waters in open oceans are now relying upon automation and subsea functions originally placed on platforms, whether bottom-fixed or floaters. Subsea separation and compression units have been recently installed and are in service. Each unit is an independent sub-system, interacting with the subsea flowline network. A comprehensive digital twin of the subsea infrastructure must include models of all these components, with functions that are inter-dependent (Al-Rbeavi, 2023).

There is currently no widespread adoption of digital twins for HPHT flowlines, and in general for subsea pipelines. However, the topic is gaining traction, led by the spreading of digital tools and concepts to all aspects of asset management, with several examples available in the literature. An extensive overview of the recent opportunities and challenges of digital twins for subsea pipelines is presented in (Chen, Videiro and Guedes Soares, 2022), with an overview of recent advancements enabling subsea monitoring presented in (Stewart, Ayaz, Fough and Prabhu, 2025). The topic of using digital twins for subsea fields starting already from the design stage is the focus of (Bhowmik, Noiray and Naik, 2019) and (Patel, Cai, Noiray and Bhowmik, 2021), without however detailing how to then extend their use in operation, which is more common to find in the literature, especially for pipelines. A theoretical framework for how to implement digital twins for pipelines in operation is e.g. presented in (Bhowmik, 2019), while (Chiamaka Ezech and Chuka Oriaku, 2021) and (Bhowmik, 2021) focus on the topic of corrosion management via digital twins. Sensor-based structural monodiagnostics subsea pipeline is discussed e.g. in (Jianxing Yu, Zihang Jin, Yang Yu et al. 2025), (Gao and Tian, 2025), with a focus on HPHT in (Hendawy, 2017) and examples of predictive diagnostic in (Fan Chen, Haotian Wei, Jiamin Tang et al. 2025). A notable real-world application is the deployment of a structural digital twin for HPHT flowlines in the Gulf of Mexico by Clarus Subsea Integrity (Acteon) for LLOG. This system monitors thermal fatigue, stress, expansion, and buckling, and includes a cloud-based portal for integrity management. Finally, while it is paramount to keep in mind that while the digital twin aims at handling uncertainties in parameters, it also has to deal with the uncertainties in the data it uses. This is explored in (Burg Demay, Donatelli et al. 2025), with DNV-RP-0497 providing methods for ensuring sufficient data quality of data.

From pipeline integrity management to dedicated software platforms

Pipeline Integrity Management Systems (PIMS) emerged decades ago as a response to the need for continuous, structured oversight of pipeline condition and safety. Originally developed for long-distance offshore gas pipelines – particularly during the transcontinental gas expansion of the 1980s – PIMS was designed to monitor flow dynamics in real time, manage data from periodic surveys, and trigger intervention protocols in the event of leaks or mechanical anomalies. These early platforms, though technologically limited compared to today's standards, represented the first integration of software into asset integrity management and served as precursors to the modern concept of a digital twin. The initial assumptions behind PIMS were rooted in relatively stable and predictable conditions, such as the uniformity of fluid composition, low corrosion risk, and the reliability of steel behavior in elastic regimes. Over time, PIMS evolved from a basic inspection and reporting framework into a more comprehensive system incorporating

data analytics, inspection scheduling, risk-based prioritization, and event simulation (Goodfellow and Jonsson, 2015).

Early subsea developments. In the early stages of offshore oil and gas development – particularly from the 1970s through the 1990s – subsea fields were relatively simple, localized systems typically consisting of a small number of wells tied back to a fixed platform or Floating Production System (FPS) via short flowlines and risers. These developments were generally situated in shallow waters, often less than 300 meters deep, allowing divers to perform installation and inspection tasks. Flow conditions were mild, with moderate pressures and temperatures, and the transported fluids were usually single-phase and non-aggressive. As a result, corrosion rates were low and the risk of structural issues like in-service buckling was minimal. Flow assurance was managed through simple measures such as insulation or routine pigging, and integrity monitoring relied on periodic inspections and basic SCADA-based systems (Y. Bai and Q. Bai, 2010). The ability to manage the integrity of the asset over the lifecycle was increasingly efficient. Only operational flexibility limited, as systems were designed around static operating conditions with conservative assumptions. These assumptions might be no longer adequate for modern subsea pipeline networks, particularly those transporting HPHT (High-Pressure High-Temperature) and chemically aggressive products. The uniformity of fluid composition, low corrosion risk, the reliability of steel behavior in elastic regimes are no longer stable and predictable, particularly in deep water field developments.

Modern subsea complexity. The picture has transformed radically. There was a full integration of design for installation and for operation, the life cycle drives the engineering flow from the earliest, accounting for new technologies leading to unmanned offshore operations is a must. This has already started in traditional districts, where production plants are on bottom-fixed platforms. Fig. 4 shows how modern inspection of subsea is approached, using new vision technology as well as remote processing of outcomes to make decisions. Modern subsea fields have evolved into complex, decentralized networks that can span tens or even hundreds of kilometers, often operating in deep or ultra-deep waters exceeding 1500 meters. These developments require the use of remotely operated vehicles (ROVs) and autonomous underwater vehicles (AUVs) for all subsea installation, inspection, and maintenance activities. They routinely handle high-pressure, high-temperature (HPHT) fluids that are frequently multiphase and chemically aggressive, leading to accelerated material degradation and more complex structural responses. In addition to basic transportation, many of these fields now feature advanced subsea processing capabilities such as separation and boosting, compression and reinjection, subsea storage and offloading, and integration with multiple reservoirs and satellite fields.

The concept of "subsea factory" has transformed the seabed into a highly automated and remotely managed industrial plant. Flow assurance has become significantly more demanding, requiring the management of risks like hydrate formation, wax deposition, scale, and slugging, often through active heating systems and sophisticated chemical treatment. Monitoring and control systems have advanced from periodic checks to real-time data acquisition supported by edge computing and AI-driven predictive analytics. These tools enable proactive integrity management and allow infrastructure to dynamically adapt to changing production profiles and uncertain reservoir behavior. As a result, today's subsea operations face much higher levels of variability and uncertainty, particularly in flow chemistry and seabed interaction, necessitating real-time condition monitoring and advanced modeling to maintain safe and efficient performance throughout the asset's life cycle.

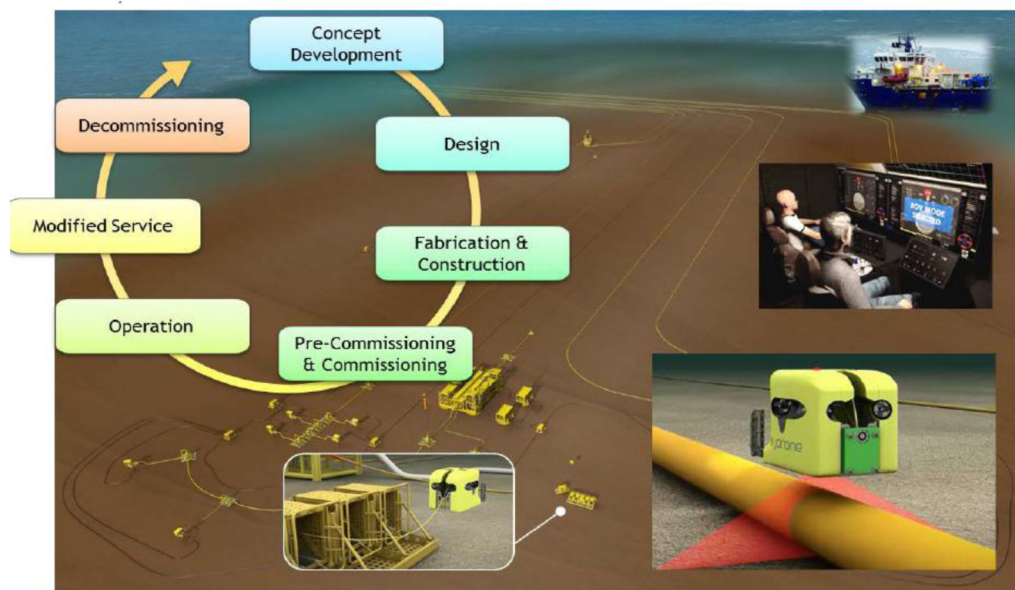


Figure 4—Autonomous underwater vehicles, vision technology and connection to onshore control tower is the new deal of underwater inspection

Challenges for flowline networks. The pipeline remains the central item of field development both in shallow and deep water. Complexity and water depth are not necessarily correlated; certain shallow-water environments impose highly active conditions that critically impact design. Across all scenarios, the thermal expansion-induced strain after startup is significantly influenced by aggressive flow chemistry and severe HPHT conditions. The material's working factor – defined in stress- or strain-based terms – is a key design parameter, generally governed by operator or regulatory requirements. The pipe concept (including material selection, welds, coatings, insulation, and corrosion protection) directly influences the flowline's carrying capacity. Corrosion presents a major threat to both internal pressure containment and resistance to external forces. Current projects, regardless of water depth, often involve highly complex subsea layouts. In ultra-deep waters fields, particular attention must be given to the cyclic or progressive development of thermal expansion recovery of the inter-field lines subject to fluctuations of high pressure and high temperature products from wells to manifold. And pipe ‘walking’ along seabed slopes and on soft seabed soils is a new risk. All pose serious risks to long-term structural integrity, especially coupling severe load effects on pipe steel and aggressive flow regimes wetting the internal wall.

Flow assurance studies are essential for anticipating the flow regimes – such as emulsions, slug flow, annular or stratified flow – under evolving conditions over the field's lifecycle. These regimes are determined by multiphase compositions, pressure, temperature, velocity profiles, and shear stress. Understanding flow regimes enables prediction and control of corrosion and erosion (the latter being a function of velocity and sand content). Flow assurance guides critical design decisions: pipe size and material selection, operational parameter ranges, insulation requirements, electric heating systems for transient conditions, and mechanical protection needed for installation and during operation. These studies must account for varying operational states, including routine shut-ins and slowdowns. The selection of pipe diameter is driven by flow rate requirements at each field location – from wellheads to manifolds or riser bases – while material selection stems from corrosion analysis based on fluid composition and service conditions. On this basis, pipe material and minimum wall thickness requirements for internal pressure containment are established.

Internal corrosion remains a non-negligible failure risk, often due to unanticipated conditions – such as the presence of sulfate-reducing bacteria in waters used for pre-commissioning or in anoxic sediments, the biological activity of which may threaten the integrity of pipe wall in service. Carbon steel (CS) remains

the material of choice due to its cost and availability. Corrosion-resistant alloys (CRAs) are considered only when CS proves inadequate for the specific flow chemistry and service conditions. Though costlier, CRA options (whether full CRA or clad – mechanically or metallurgically bonded) may be the only technically viable solution. Therefore, material selection must involve rigorous analysis to ensure both technical performance and economic feasibility.

In-service buckling due to thermal expansion is a critical concern for HPHT flowlines in deepwater environments where pipelines are exposed and subject to flow fluctuations, shutdowns, and restarts. To manage this, modern design incorporates lateral buckling triggers at calculated intervals, to guide thermal strain recovery and prevent uncontrolled deformation. Robust design requires accounting for pipeline-soil interaction and seabed variability. Due to inherent uncertainties, the consequences of potential buckling must be evaluated early in design. Even for conservative design assumptions, when in operation, the actual pipeline behavior may diverge from predictions. Thus, design efforts focus on enabling pipelines to tolerate maximum anticipated stress and strain with sufficient safety margins.

Subsea integrity management, challenges from design onward. Field development involves managing a complex system of assets in a confined, congested area. Each component has distinct operational requirements, yet all must function cohesively. High environmental sensitivity demands that loss of containment be prevented throughout the entire field lifecycle – from design and fabrication to commissioning and operation. Comprehensive analysis, survey programs and advanced monitoring tools are essential, particularly under severe conditions with high failure consequences. Integrity assessments are multidisciplinary, including product and transport condition analysis, flow assurance modeling, corrosion mechanism identification, material loss rate prediction, thermal expansion response analysis, weld integrity evaluation, and defect tolerance assessments. The working factor of the line pipe material in operation, whether after stress or strain-based design criteria, is always a discriminant, generally ruled by company or authority requirements (R. Bruschi et al. 2017). Traditionally, subsea pipeline design was based on a safe-life philosophy, relying on over-engineering with generous safety margins, thick walls, and corrosion allowances to compensate for uncertainties. Historically, integrity management relied on routine inspections scheduled at fixed intervals, with criteria based largely on initial design assumptions and minimal operational feedback. Today, design practices have shifted toward a fail-safe and adaptive approach, which anticipates degradation and incorporates in-service monitoring and risk-based interventions. Modern solutions are tailored to specific field conditions, supported by integrated digital design tools, including coupled flow-structural simulations and 4D reservoir modeling. Modern integrity management is driven by real-time condition monitoring of operation parameters. Autonomous underwater vehicles (AUVs) now perform inspections using advanced diagnostics, and decision-making is increasingly risk-informed, supported by integrated simulation models and probabilistic analysis to proactively manage asset integrity throughout its life cycle. Cost effective solutions shall be based on a rational combination of initial robustness accompanied by the ability to intervene after having detected mishaps by monitoring the field performance.

Modern field developments often consist of large, interconnected networks of flowlines with shared manifolds, riser bases, and floating production units. Operational flexibility, process transients, and varying reservoir conditions make past design approaches ineffective in ensuring long-term integrity. Under HPHT conditions, pipeline response to thermal expansion must be carefully controlled; otherwise, global buckling or progressive pipe walking can initiate structural degradation. Additionally, the localized and time-dependent nature of corrosion under multiphase flow conditions renders traditional inspection planning inadequate. These conditions have necessitated a new paradigm: moving from a safe life to a fail-safe approach. Rather than relying solely on overdesign, engineers now implement condition monitoring strategies integrated into real-time platforms – enabled by the digital twin. These modern digital twins extend the legacy of PIMS by including real-time data, predictive analytics, probabilistic design models, and

simulation tools that evolve with the asset throughout its lifecycle. The convergence of sensor technology, edge computing, data analytics, and frameworks like DNV-RP-A204 has enabled digital twins to evolve into powerful decision-support systems – capable of simulating complex behaviors and informing human or automated interventions. What distinguishes today's approach is the shift from periodic inspection and deterministic safety margins to continuous condition monitoring, enabled by automated data acquisition, edge computing, and high-fidelity simulation models. The digital twin is no longer simply an advanced version of a PIMS. Instead, it should allow for a dynamic, real-time functional and structural representation of the pipeline network, continuously updated with live data and used not only to detect anomalies but also to anticipate them through predictive analytics. The role of a digital twin is not limited to individual pipelines but extends to the network. A systemic digital replica can account for the interaction between pipelines, real-time variations in flow allocation, and the response of structural components to shared loads and thermal gradients.

Despite all advancements, the transition remains incomplete. Many subsea developments still rely on PIMS-style platforms that lack functional integration or real-time adaptability. Barriers include regulatory inertia, legacy design philosophies, and the scarcity of multidisciplinary professionals capable of managing hybrid platforms combining structural mechanics, fluid dynamics, and data science.

Nevertheless, the direction is clear. As field complexity increases and lifecycle performance becomes a central economic and safety concern, digital twins are emerging as not only viable but necessary tools in the design and operation of HPHT subsea flowline networks. What began as software tools for basic monitoring have now matured into intelligent, condition-aware systems that embody the next generation of pipeline integrity management.

Functional replica and structural health monitoring of HPHT pipelines

Safe-life design philosophies imply conservative material selection, thick wall sections, and oversizing of corrosion allowances – all with the hope of avoiding any need for mid-life interventions. However, this philosophy can be prohibitively expensive and may not even guarantee safe long-term service, especially under HPHT or corrosive conditions. Moreover, it provides little flexibility in adapting to real-life operating conditions, which often differs from initial design assumptions. Instead, by moving toward a fail-safe design, enabled by continuous condition monitoring and rational, risk-informed intervention protocols, operators gain the flexibility to optimize operations and extend asset life without committing to overly conservative initial design strategies. The future of subsea development will likely rely not on further physical overdesign, but on data-centric and adaptive engineering frameworks that continuously reconcile real-world behavior with predictive models. This is not merely an evolution in tools, but a fundamental shift in philosophy – from static to dynamic, from safe-life to fail-safe, from periodic inspection to continuous insight.

DT of HPHT flowline infrastructure shall tackle the three main challenges: corrosion, flow assurance, structural integrity. DNV-RP-A204 provides the methodology: the first step is dividing the systems into manageable parts, named functional elements (FEs), i.e. modular components that support specific decision-making tasks – see Fig. 5. Each FE typically replicates a key functionality of the asset. 3 FEs are then identified: Three FEs are identified:

- Flow Assurance Modelling and Monitoring (FAMM-FE), monitoring physical and chemical parameters, feeding them into a multiphase flow model to assess performance and detect deviations or degradation trends;
- Corrosion Monitoring and Management (CMM-FE), monitoring the corrosion state of the flowline and degradation trends;
- Structural Integrity Monitoring and Management (SIMM-FE), monitoring the load effects within the flowline vs strength degradation due to corrosion.

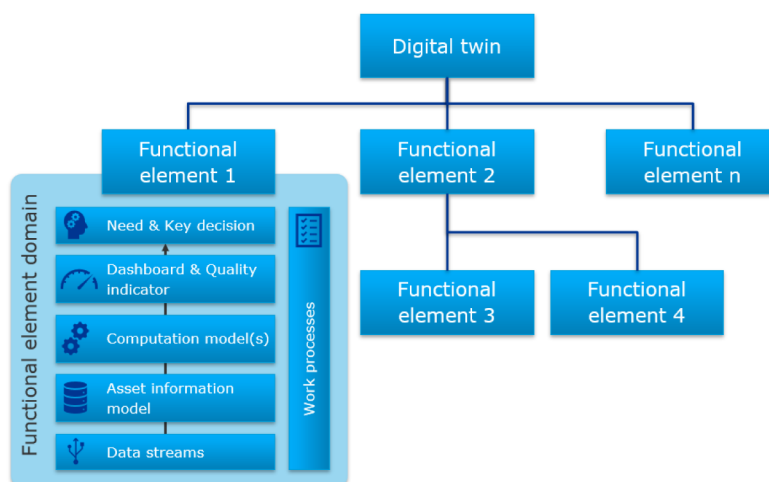


Figure 5—Digital twin as a collection of functional elements, FE (from DNV-RP-A204).
In case of field development projects, the list of functional elements is increasingly long for increasing water depths and distances from traditional offshore OG districts.

FEs are fed by data coming from sensors, that are used by numerical simulation models for engineering prediction, in combination with asset databases, to register the condition of the flowlines over time. One single flowline at a time is considered, as different pipe concepts and concerns often discriminate against each other. Then, the approach is extended to the full flowlines system, noting that the integrity of each flowline always relates to the optimization of global field production. The FEs are used to confirm the design assumptions and to provide early alarms in case of deviations. The goal is to leverage field data in combination with modelling to assess the present condition and to get the best prediction of progress in time. As per Fig. 5, there are 6 items that define an FE. The key need is what leads to the development of the FE in the first place. The remaining items are explored below.

Data

The foundations of FEs are data streams, which feed the computational models. Which data needs to be collected obviously depends on the computational model of the considered FE and therefore varies depending on the implementation. Some general consideration can be given:

- FAMM-FE: the data are related to the chemical and physical properties of the fluid, namely composition (water cut, gas volume fraction, etc.), temperature, pressure, density, viscosity, flowrate. These can be measured at the inlet and outlet of the flowline section. Data are used to feed flow assurance models, to predict current flow regime as an input to corrosion assessment.
- CMM-FE: the data are discretely collected from pig runs, planned to provide indications of corrosion entity and patterns, often coupled with measurements at purpose coupons or outlet filters to get a measure of global corrosion, to be compared with outcomes of corrosion type and rate from models fed by the findings of flow assurance calculations.
- SIMM-FE: the data can be obtained in different ways, from measuring x-y-z along the flowline, using geo-pig or external surveys, to get the shape of the route and of the critical curvatures; to direct measures of the strains along the flowline or at critical locations, located to capture the full structural behavior. x-y-z data is used to update the structural models based on as-laid or after-commissioning conditions.

It is paramount to ensure that data is provided with sufficient quantity and quality (see e.g. DNV-RP-0497). Unfortunately, real time corrosion data as well as stress and strain along the flowline are hard to

get. It is necessary to rely upon advanced modelling, fed by available data purpose planned in time and at specific locations, to provide a rational picture of the condition.

Despite the subsea challenges, technology is mature to allow for a direct and continuous measurement of a few parameters. However, the actual need for real time measurements shall be evaluated. For HPHT flowlines, transporting aggressive fluids, this is paramount for flow assurance. Corrosion, instead, is a slow process that can be measured at regular intervals, established at design stage and reconsidered in service. Finally, the state of stress might in principle be measured real time, however related sensors are not fully proven for subsea. Two are the real needs for structural integrity: measuring the movements of the flowline on the seabed due to thermal expansion and flow fluctuations, to assess curvature localization; and measuring the strength capacity (bursting, local buckling, oligo-cyclic fatigue) along the flowline, as jeopardized by corrosion. The following is then needed:

- real time measurements of the flow and its properties;
- magnetic flux and/ultrasonic pig at regular intervals, e.g. 5 years, for assessing the pipe wall corrosion;
- geo-pigging and external surveys to assess x-y-z profile of the flowline, for assessing bending strains and lateral migration.

Data that are gathered during regular inspection requires dedicated ancillaries; real time data requires tools for which the performance in the long run is a must, for which replacement in case of misfit is possible.

Sensors. The sensor system shall be integrated into the DT data platform and assured as being able to provide the necessary reliability of the measured properties. Guidance on minimum requirements for sensor systems can be found in DNV-RP-0317.

- For flow assurance, chemical analysis of flow composition is challenging. Most sensors measure bulk properties (water fraction, density) rather than full molecular breakdown. For the latter, R&D regarding subsea analyzers is ongoing, based on microfluidics and lab-on-chip technology for advanced chemical analysis on fluid samples (Monacchi et al., 2021). For the former, fully fledged laboratory instruments are uncommon on the seafloor, but specialized in-line sensors can infer composition. Gas chromatographs and optical spectrometers can e.g. be used to determine the concentrations of various components (hydrocarbons, water, gas fractions). There are also electrochemical sensors for specific species (like H₂S or pH sensors for water content), though normally used in downhole or topside. Moreover, multiphase flow meters, incorporating composition sensing, can also be used.

Flow rate is measured by flow meters; for single-phase fluids, common flow meters include differential pressure and ultrasonic flow meters. Turbine flow meters and electromagnetic flow meters for conductive fluids are used on topside. For multiphase flows, specialized multiphase flow meters are used, that combine multiple sensor inputs to deduce the flow rates of each phase, possibly using cross-correlation technologies, where sensors detect flow perturbations at two locations and correlate them to calculate flow velocity. Newer designs use acoustic sensing of turbulent eddies on the pipe wall to derive flow.

Temperature is commonly measured with standard industrial sensors adapted for external pressure due to water depth, such as thermocouples and resistance temperature detectors, built into robust probes and placed at critical locations. Distributed temperature sensing use fiber-optic cables and laser: the fiber can measure temperature along the entire length of the flowline, by sensing temperature-dependent backscatter. Fiber-optic is ideal for leak detection, by profiling the temperature along the flowline, being immune to electromagnetic interference and able to cover long distances.

Pressure measurement is an established technology; in subsea conditions, sensors are typically strain-gauge or piezoelectric-based transducers, packaged in high-pressure enclosures.

- For corrosion, rates and patterns are hardly measurable in real time. Based on experience on surface plants, the technology based on coupons can be considered real time but provides only generic data about flow aggressivity, lacking indications on corrosion shape and distribution along the flow line. Corrosion is then measured by performing pig runs, with sensor probes based on magnetic-flux or ultra-sonic technology, to measure loss of metal and cracks, respectively. Detection cycles between two pig runs account for an anticipated corrosion rate, established at design stage and possibly modified during operation. Direct corrosion measurements can also be performed through wall thickness sensors installed at selected locations to directly measure pipe wall thickness.
- For structural integrity, the configuration of the flowline along the route, which might change in service due to HPHT fluctuations (in service buckling, pipe walking), need to be measured. The change on the seabed is discontinuous, linked to flow fluctuations, geohazards, and external mechanical interference. Flowline sections might also be exposed and suspended over seabed, e.g. free spanning or jumpers, with continuous loading from passing flow slugs or external cross flows causing vortex induced vibrations. In this case, the reference limit state is fatigue, needing real time measurement of vibrations. The primary vibration sensor is the accelerometer, often piezoelectric. Subsea accelerometers are housed in waterproof enclosures and can be bolted or clamped to the flowline. There are also acoustic-based systems, hydrophones or acoustic pressure sensors that measure the sound emitted by the vibrating structure.

Environmental loads also need to be measured. In combination, wave radars and acoustic doppler current profiler are used to collect data to reproduce near-seabed hydraulics that impact the flowlines. They are both well established in industry.

The interest in this paper is for the impact of HPHT fluctuations on the configuration of the pipe on the seabed, particularly localization of deformation at critical locations. Commonly, deformation is measured via strain sensors, either traditional foil strain gauges bonded to the structure, or modern fiber-optic strain sensors (fiber Bragg gratings or interferometric sensors) bonded or clamped to the pipeline. It is however hard to successfully install strain gauges on subsea flowlines (bonding, protection from water ingress, durability). Displacement can also be measured directly, e.g. using linear variable differential transformers (or extensometers) to measure relative movement or gap opening at joints supports. For flowlines in HPHT service, interest is to get x-y-z data that allow for getting a measure of curvatures (in service buckling) and displacements (pipe walking). In both cases, triggering events are flow fluctuations, followed by slow adjustments. Internal surveys, using geo-pig, can provide the correct alignment of flowline, possible coupled with corrosion measurement via multifunction pigs. External survey, using modern tools mounted on underwater vehicles coupled with transducers clamped on the flowline, can also provide data that allow to provide the alignment sheet of the flowline.

Sensor systems will need to be integrated in the DT data platform and be assured as being able to provide the necessary reliability of the measured properties (see DNV-RP-0317).

Asset information model

Ensuring that asset properties and data are consistently defined throughout the project disciplines and along the asset lifetime is a central element for any DT that wants to cover the entire asset lifecycle. This is achieved by defining consistent asset information model (AIM) for the system. The purpose of the AIM is to provide structure and context to the data, to enable data exploration (search, navigation, merging). Moreover, the AIM can be used for the validation of data and change management. The AIM can include all sorts of data, such as data streams, metadata, relationships between data, master data, spatial data, computation models,

documentation. According to DNV-RP-A204, AIMs can have different levels of complexity (detached, basic, integrated, holistic). An integrated AIM is necessary for the digital replicas of the HPHT flowlines. Data are from multiple sources, so they need to be structured, integrated, accompanied by relevant metadata. Moreover, foreign keys (tags) and master data need to be managed through the AIM. AIM is the evolution of PIM, commonly regarding a single pipeline. AIM is linked to computational models, the core of each FE, where field data are processed to fine tune the digital replica of functions and integrity of the flowlines.

Computational models

Computational models can be physics-based or data driven. For flowlines, physics-based models are for flow assurance and for structural integrity. They are developed at design stage, using data from design basis and fundamental principles. Field data are then used to fine tune the model, with the aim of closely mirroring physics. Data-driven models bypass the need of a detailed understanding of the physical process, but they need high-quality data, many and representative, a challenge for real-world systems (Al-Ammari et al. 2025). Hybrid approaches, which fuse physics-based insights with data-driven adaptability, are increasingly popular (Ezeh and Oriaku, 2021). They combine the understanding of the physical phenomenon with the flexibility of data-driven engineering models.

- The FAMM-FE incorporates flow simulation that mirrors the actual flowline. This typically involves a 1D transient multiphase flow model, integrated into the subsea network (e.g. Olga software). The model is based on conservation of mass, momentum, energy, and rigorous thermodynamics. From these, the flow regimes from inlet to outlet are represented, acting as a real-time virtual flow simulator that runs in parallel with the actual pipeline. The relevant parameters are originally from design basis, then are purpose calibrated to each flowline. Friction factors, heat transfer coefficients, and equipment in line details are fine-tuned so that the model's predictions align with performance, e.g. matching historical pressure drop data. The model computes pressure profiles and flow velocities, specifically predicts the flow regimes (e.g. stratified, annular, slug flow) along the pipeline length under the given conditions. The flow regime is the main input to corrosion prediction tool. In deep offshore settings, the model continuously checks the local temperature / pressure against hydrate formation curves. If hydrates are likely, the twin can suggest mitigation actions (e.g. injecting methanol or glycol inhibitors). Through the model, anomalous conditions can be monitored: when developing slug flow (characterized by large oscillations in flow and pressure), it can set an alert to adjust choke settings or ramp up a separator accordingly. Or an expected cooldown during shutdown prompts pre-emptive inhibitor injection to avoid hydrate plugging.
- In the CMM-FE, flow regime and relevant parameters from the FAMM-FE are used as input, combined with the measurements obtained through the regular inspection campaigns. The main corrosion processes are sour corrosion and sweet corrosion, depending on the fractions of CO₂ and H₂S. At design stage, careful predictions of production flows and flow regimes address the most probable corrosion mode and define the expected rate. However, capturing complex nonlinear relationships regarding corrosion rate and shapes is difficult with a physics based model. In service then, a data driven model for corrosion can be used (Chiamaka Ezeh and Chuka Oriaku 2021), using historical data to train the model, and FAMM-FE and inspection results to make predictions. The first step is to extract features from the historical data (typical features would be temperature, pH, flow rate, CO₂ concentration, pipe age, previous failures, etc.). Once features are defined, supervised learning algorithms are trained to recognize corrosion patterns. Random forests, gradient boosting machines, and support vector machines techniques can be then used for classifying pipeline segments by risk level or for forecasting corrosion rates under specific operating conditions. These models learn from labeled datasets, where outcomes like "low

corrosion risk", "moderate corrosion risk", or "severe corrosion" are predefined based on inspection data. Over time, corrosion predictions are reviewed, and new models regarding the specific flowline are trained by using updated in line inspection records and mitigation (injected fluids) logs, if any. Methods to ensure that models are trustable are defined in DNV-RP-0513, as regards physics-based models, in DNV-RP-0510 as regards data-driven applications, in DNV-RP-0665 in machine learning applications for data-driven models.

- The physical parameters from the FAMM-FE and the corrosion patterns along the pipeline from the CMM-FE are input for the SIMM-FE. The structural models covers the flowlines, the termination ends (PLET), the jumpers connecting PLET and manifolds, the manifold. The flowlines are mainly subject to flow fluctuations, that might cause lateral and vertical deflections to recover the thermal expansion. Except lengths in suspension due to seabed features, where flow induced oscillations (both external due to VIV, and internal, due to high-speed slug flows), they are assumed as static structures, for which alteration of as-laid settlement is only due to thermal recovery. The same applies to PLET and manifold. Jumpers are similar to suspended flowline lengths. Further, subsea built at the toe of continental slopes might be subjected to geohazards. Nonetheless, structural design models (commercial software, like ABAQUS and ANSYS or similar) are well proven, provided input parameters cover variability during service, and safety margins address random and epistemic uncertainties. Monitoring is requested to verify the flowline alignments, that can be altered in consequence of flow oscillations, predicted or unpredicted at design stage. The FAMM-FE can be fed with field data, to get an assessment of material usage at critical locations (local buckling limit state). On top of this, the corrosion patterns from measurements are applied to the pipe wall, to update the measure of safety margins with respect to leak or bursting limit state. Moving toward a fail-safe design, enabled by continuous condition monitoring and rational, risk-informed intervention protocols, operators gain the flexibility to optimize operations and extend asset life without committing to overly conservative initial design strategies.

Practically, physics-based and data-driven models interplay, exchanging input-output at first between FAMM-FE and CMM-FE, then between FAMM-FE and SIMM-FE, eventually between CMM-FE and SIMM-FE. It is fundamental to ensure that the models can be trusted and represent the actual condition of the system over its lifetime. Methods to assure that models are trustable are defined in DNV-RP-0513 for physics-based models, and in DNV-RP-0510 and DNV-RP-0665 for data-driven models.

Data quality indicator

This is a fundamental piece of a DT, according to DNV-RP-A204. It is used to continuously inform the user about the condition of the FE and DT. The indicator includes continuous assessment of the quality of input data and the results from the computation models. As a reference, the potential failure modes of the physical system with respect to the outcomes of the DT infrastructure. In particular, the changes of the physical system after warning from the digital replica shall be always processed with care, in relation with the performance of the computation models and data quality, both impacting DT's trustworthiness. The continuous assessment is typically automated, while the periodic assessment is typically performed upon request. An indicator can be developed as a 'traffic light' system, with green meaning that the FE can be trusted, yellow that something odd is going on and red that the system results should not be trusted.

Working procedures

Traditional approach implies conservative material selection, thick wall sections, and oversizing of corrosion allowances – all with the hope of avoiding any need for mid-life interventions. It can be prohibitively expensive and may not even guarantee safe long-term service, especially under HPHT or corrosive conditions. The new approach provides flexibility in adapting to real-life operating conditions,

which often differs from initial design assumptions. For realizing the advantages, the digital twin must be integrated in the company's work processes. Specifically, the outcome of the monitoring is used as input in a risk-based inspection strategy, where inspection intervals are defined based on the risk of corrosion or of structural integrity degradation. If the risk is low based on the predicted corrosion rate and structural condition, inspection intervals can be extended.

Conclusions

The focus of this paper is on the digital replica of the HPHT inter-field pipeline network, where production fluids transit from well-heads to subsea manifolds, then reach the riser base or export pipeline. Such fluids might be aggressive, imposing impacting choices at design stage, that might be not enough during the life cycle. The increasing complexity of subsea network is currently due to new components, increasingly present in new subsea, at the cutting edge of subsea technology, such as flow separators and subsea boosting/compression stations. This is only part of the overall complexity of field development infrastructure, the planning of which for a satisfactory performance in operation shall necessarily start at the earliest design stage.

Dealing with complexity of offshore field development

The development of digital replica of field development infrastructures, from components through sub-systems to inter-connectivity of each one to attain the performing 'orchestra', might be an answer. This is nothing new at design stage, the subject is the software platform that assists and facilitates traditional design of sub-systems and their integration to get the system. The novelty is only that such platform evolves in operation, using real data from field, that allows the digital replica to mirror at best the real environment. So, where there is complexity, optimization and performance improvement can be pursued through the development of a digital twin. Complexity justifies the implementation of the novel approach. As regards the design for operation of HPHT inter-field pipeline network, it is reckoned that design is affected by uncertainties, both random input parameters and episteme or capability to predict performance through engineering models. This contributes to evolving the traditional safe life design into a data-driven, fail-safe approach for a safe and cost-effective design for operations. DT can be the main actor.

Traditional vs advanced design of HPHT flowline networks

Traditional design aims at managing project risks and uncertainties through rational choices, based on data, methods, criteria, and safety margins. This might imply high investment costs, but it requires little care during operation, provided field conditions are and remain consistent with early design assumptions, which might not be the case. On the other hand, fail-safe design may lead to design choices that are less cost impacting, but it relieves continuous monitoring via a digital twin during operation. The aim of fail-safe design remains on the one hand to ensure satisfactory performance and safety margins between time milestones during operation, on the other hand and in case of unacceptable trend at established check-up milestones, to allow for deploying additional remedies to manage conditions that might develop into misfit in service. Further, differences with respect to original assumptions might impact on the safe service. Mitigation measures, belonging to preparedness plan, can be soon deployed.

Project experience of HPHT inter-field pipelines shows that two are the concerns that need to be addressed: 1. the chemical composition, the fluid composition, the flow regimes, and their variability during the asset lifespan are uncertainties difficult to solve at the earliest design stage, and which strongly affect the pipe concept and design parameters. 2. Severe HPHT flows and seabed conditions for the as-laid pipeline affect the response under flow fluctuations. The management of the concerns at design stage may lead to different results: flow assurance to including thermal insulation (at worst, electric heating of the pipe wall) or pipe burial and cover; aggressive fluids to a corrosion management approach, with inhibitors and regular pigging; loss of metal due to corrosion to generous wall thickness allowance or selection expensive alloys;

severe HPHT and flow fluctuations to purpose developed equipment and procedures to control the load effects from thermal expansion. As an example, in case of sweet corrosion, in a traditional design approach, the wall thickness allowance is based on conservative estimation of wall thickness loss; this implies either thick walls or cladding with a corrosion resistant alloy, the latter with the aim of limiting the implication from pipe weight. But corrosion prediction is uncertain. In a fail-safe approach, flow assurance and corrosion monitoring can support the control of corrosion development, providing on the one hand a rational limit to corrosion allowance on the other appealing to the control of corrosion over the life cycle, with measurements and remedies (inhibitors, flow management) only in case of excessive rates or new flow properties.

Benefits from digital twin

The role of a digital twin of the interfiled pipeline network is described by looking at the performance of a single carrier pipe, examining the functional elements that compose the proposed digital twin, the data that needs to be acquired, the sensor system that needs to be used, the computational models that should be developed. What occurs on one flowline shall be extended to the whole network. This is the real complexity, linked to production strategy and well-head control system, as flow assurance is the origin of all concerns. For purpose, digital twin is not a completely new concept, but rather an evolution and combination of already existing and proven methods. Flow assurance and in service buckling design are disciplines deeply involved in field development projects, with low and high-fidelity software that can predict the system performance. By combining both on the software platform, for each flowline of subsea network, feeding the model with real flow parameters and response monitoring, it can provide a confident prediction, as well as the flexibility to change flow rates and their physical path. DT can be used to mitigate the uncertainties related to the design assumptions and to the computational models, implementing an effective condition monitoring system. The coupling of multi-physics simulation models with field data is the digital twins. In principle, this should not be a big change as regards modelling, with the following recommendations: the digital twin for the specific field development should be developed already at the earliest design stage; during design, the digital twin should be leveraged continuously until installation, considering commonly changing data regarding the management of flow assurance, as a function of outcomes of 4D simulation of reservoir exploitation, that are continuously run; the digital twin model must be updated based on as-laid and as-built data regarding each flowline, as well as based data gathered during pre-commissioning; condition monitoring is implemented during operation based on the regular flow of data from installed instrumentation, coupled with survey data, regular or extraordinary, performed by ROV or AUT, likely resident in remote offshore districts; modifications of remotely actuated valves and subsea control boxes, controlling the production and actuation flows, shall be included.

New concerns

The digital twin is an old concept that is now enabled by new technologies, with potential to greatly benefit to design for operation of complex infrastructures for field development. For subsea intra-field network, only a small but pivotal sub-system of complex infrastructures, expensive design choices might not be safe as assumed during the life cycle, so condition monitoring can contribute to managing misfits. The question of traditional operators, should DT be a novel and expensive complexity, has a twofold answer: software and hardware are increasingly less expensive; difficulty comes from the need for new skills, hardly available as few and engaged in competitive business. The latter is main resistance to a change of paradigm.

Acknowledgement

The authors would like to express gratitude to Det Norske Veritas (DNV) for supporting the paper, outcome of international developments regarding new concepts of digital twins for field development projects. DNV, involved in the earliest examples of digital twins for the maritime industry, has addressed the present investigation with internal documents and recommended practices.

Bibliography

- Al-Ammari, W. A., Sleiti, A.K., Rahman, M.A. et al 2025. Digital twin for leak detection and fault diagnostics in gas pipelines: A systematic review, model development, and case study. *Alexandria Engineering Journal*, Volume 123. <https://doi.org/10.1016/j.aej.2025.03.054>.
- Amaechi C.V., Hosie G., Reda A. (2023): "Review on Subsea Pipeline Integrity Management: an Operator's Perspective"; *Energy* 2023,16, 98
- Augustyn D., Tygesen U. T., Ulriksen M.D., Sørensen J.D. (2019): "Data Driven Design and Operation of Offshore Wind Structures"; Proceedings of the Twenty-Ninth ISOPE Conference, Honolulu, Hawaii-USA
- Bai, Y. and Bai, Q. 2010. *Chapter 12 – Subsea System Engineering*. Subsea Engineering Handbook, 2010 Elsevier Inc.
- Bhowmik, S. 2019. Digital Twin of Subsea Pipelines: Conceptual Design Integrating IoT, Machine Learning and Data Analytics. Paper presented at the Offshore Technology Conference, Houston, Texas, May 2019. <https://doi.org/10.4043/29455-MS>.
- Bhowmik, S. 2021. Digital Twin for Offshore Pipeline Corrosion Monitoring: a Deep Learning Approach. Paper presented at the Offshore Technology Conference, Virtual and Houston, Texas, August. <https://doi.org/10.4043/31296-MS>.
- Bhowmik, S., Noiray, G. and Naik, H. 2019. Subsea Pipeline Design Automation Using Digital Field Twin. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November. <https://doi.org/10.2118/197394-MS>.
- Bruschi, R., Bartolini, L.M., Cherubini, P. et al 2017. Meeting the challenges of upcoming subsea field developments: from carbon steel line pipe to new materials and pipe concepts. Proceedings of the 27th International Ocean and Polar Engineering Conference, San Francisco, CA, USA, June 25-30.
- Burg Demay, M., Donatelli, G., Meira de Oliveira, A.L. et al 2025. On the influence of uncertainty in risk analysis based on digital models. *Measurement: Sensors*, Volume 38, <https://doi.org/10.1016/j.measen.2024.101510>.
- Chen, B.-Q., Videiro, P. M., Guedes Soares, C. 2022. Opportunities and Challenges to Develop Digital Twins for Subsea Pipelines. *Journal of Marine Science and Engineering* 10, no. 6: 739. <https://doi.org/10.3390/jmse10060739>
- Chen, F., Wei, H., Tang, J. et al 2025. Digital twin based predictive diagnosis approach for submarine suspended pipelines. *International Journal of Pressure Vessels and Piping*, Volume 214. <https://doi.org/10.1016/j.ijpvp.2025.105451>.
- Chiamaka Ezech and C., Chuka Oriaku, C. 2021 Digital Twin Technology: a Novel Approach to Pipeline Corrosion Monitoring. *International Journal of Computer Applications Technology and Research (IJCATR)*, Volume 10, Issue 12. <https://doi.org/10.7753/IJCATR1012.1006>.
- DNV-RP-A204, Assurance of Digital Twins. 2023
- DNV-RP-0317, Assurance of data collection and transmission in sensor systems. 2023
- DNV-RP-0497, Assurance of Data Quality Management. 2023
- DNV-RP-0510, Assurance of data-driven applications. 2024
- DNV-RP-0513, Assurance of simulation models. 2021
- DNV-RP-0665, Assurance of machine learning applications for data-driven models. 2023
- Gao, R. and Tian, H. 2025, Multi-fidelity strength monitoring method for dynamic response of deep-sea pipelines based on digital-twin technology, *Applied Ocean Research*, Volume 154. <https://doi.org/10.1016/j.apor.2025.104414>.
- Goodfellow, R. and Jonsson, K. 2015. Chapter 1 – Pipeline Integrity Management Systems (PIMS). *Oil and Gas Pipelines: Integrity and Safety Handbook*. Published in 2015 by John Wiley & Sons, Inc.
- Helgaker J.F., Ytrehus T. (2012): "Coupling between Continuity/Momentum and Energy Equation in 1D Gas Flow"; *Energy Procedia* 26, 2nd Trondheim-N Gas Technology Conference
- Hendawy, M. 2017. The Application of Fibre Optic Instrumentation for HPHT Pipeline Bundle Health Monitoring. Paper presented at the Offshore Technology Conference, Houston, Texas, USA, May. <https://doi.org/10.4043/27690-MS>.
- Monacchi, G., Aloigi, E. and Radicioni, A. 2021. Innovative Modular and Digitalized Subsea Sampling and Measurement System for Produced Oil and Gas. Paper presented at the OMC Med Energy Conference and Exhibition, Ravenna, Italy, September.
- NATO 2023. Role in Protecting Critical Undersea Infrastructures. Center for Strategic International Studies, CSIS Briefs, www.csis.org
- Patel, H., Cai, J., Noiray, G. et al 2021. Digital Transformation and Automation of Flow Assurance Engineering Workflows Using Digital Field Twin. Paper presented at the Offshore Technology Conference, Virtual and Houston, Texas, August. <https://doi.org/10.4043/31017-MS>.
- Rognø H. (2014): "Statoil Subsea Factory"; *Gas Technology Conference*, Trondheim-N
- Shadravan, A. and Amani, M. 2012. HPHT101-What Every Engineer or Geoscientist Should Know about High Pressure High Temperature Wells. SPE 163376
- Stewart, C., Ayaz, B., Fough, N. et al 2025. Towards the underwater internet of things for subsea oil and gas monitoring. *Internet of Things*, Volume 30. <https://doi.org/10.1016/j.iot.2025.101510>.

- Van M., Edwards C., Tran-Thanh L., Bonney M. (2023): "Digital Twins for Marine Operations: from Surface to Deep Waters"; *UK-RAS White Papers*, www.ukras.org.uk
- Yang C., Cai B., Zhang R., Zou Z., Kong X., Shao X., Liu Y., Shao H., Khan J. A. (2023): "Cross-Validation Enhanced Digital Twin Driven Fault Diagnostic Methodology for Minor Faults of Subsea Production Control System"; *Mechanical Systems and Signal Processing* **204**
- Yu, J., Jin, Z., Yu, Y. et al 2025. Deep-sea pipeline damage identification using digital twin-assisted enhanced meta-transfer learning. *Ocean Engineering*, Volume **324**. <https://doi.org/10.1016/j.oceaneng.2025.120723>.
- Zhang G., Qu H., Chen G., Zhao C., Zhang F., Yang H., Zhao Z., Ma. M. (2019): "Giant Discoveries of Oil and Gas Fields in Global Deepwaters in the Past 40 Years and the Prospect of Exploration"; *Journal of Natural Gas Geosciences* **4**